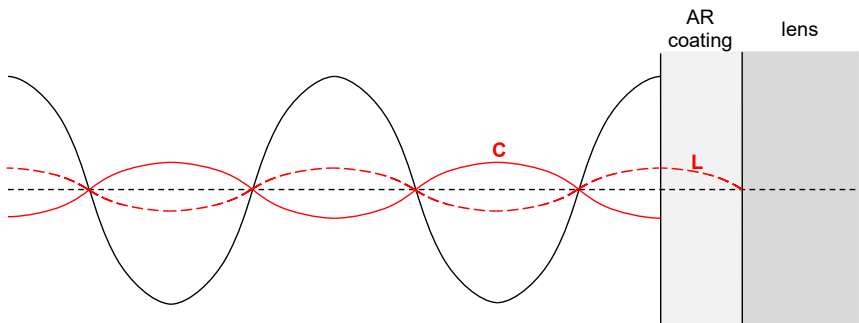


7				
	(a)	(i)	[Method 1] power = intensity $\times$ area so $R = \frac{\text{intensity of reflected wave} \times \text{area}}{\text{intensity of incident wave} \times \text{area}} = \frac{\text{intensity of reflected wave}}{\text{intensity of incident wave}}$ intensity $\propto$ amplitude² so $R = \frac{k \times (\text{reflected wave amplitude})^2}{k \times (\text{incident wave amplitude})^2} = \left(\frac{\text{reflected wave amplitude}}{\text{incident wave amplitude}} \right)^2$ since r is the ratio of amplitudes, so $R = r^2 = \left(\frac{n_2 - n_1}{n_2 + n_1} \right)^2$ [Method 2] intensity $\propto$ amplitude² $\square$ intensity = $k \square$ amplitude² power = intensity $\times$ area = $k \square$ area $\square$ amplitude² $R = \frac{k \times \text{area} \times (\text{reflected wave amplitude})^2}{k \times \text{area} \times (\text{incident wave amplitude})^2} = \left(\frac{\text{reflected wave amplitude}}{\text{incident wave amplitude}} \right)^2$ since r is the ratio of amplitudes, so $R = r^2 = \left(\frac{n_2 - n_1}{n_2 + n_1} \right)^2$	B1 B1 (B1) (B1)
		(ii)	1. sum of reflected and transmitted power (or energy) equal incident power (or energy)	B1
			2. $T + R = 1$ or $T = 1 - R$ $T = \frac{(n_2^2 + n_1^2 + 2n_2n_1) - (n_2^2 + n_1^2 - 2n_2n_1)}{(n_2 + n_1)^2}$ $T = \frac{4n_2n_1}{(n_2 + n_1)^2}$	M0 A1
		(iii)	1. at air–lens interface, $R = \left(\frac{1.5 - 1.0}{1.5 + 1.0} \right)^2 = 0.04$ and $T = 0.96$ at lens–air interface, $T' = 0.96 \square 0.96 = 0.9216$ percentage of transmitted light = $0.9216 \square 100\% \square 92\%$	C1 M1
			2. at air–lens interface, $R = \left(\frac{1.58 - 1.0}{1.58 + 1.0} \right)^2 = 0.050538$ and $T = 0.94946$ at lens–air interface, $T' = 0.94946 \square 0.94946 = 0.90147$ percentage decrease = $(1 - 0.90147) \square 100\% = 9.85\%$	C1 A1
	(b)	(i)	(from Fig. 7.4, reflection is minimised when $n_c < n_{\text{lens}}$) reflected waves from air–coating and coating–lens interfaces undergo same 180° phase change	B1

			(wave in coating travels additional half a wavelength,) both reflected waves have half a wavelength path difference	M1																		
			both reflected waves superpose destructively to give minimum/zero displacement/ amplitude	A1																		
			<i>Note: Technically the thickness of the coating is related to the refractive index. However, the question is simplified because optical path difference is not stated explicitly in H2 Physics learning outcome.</i>																			
		(ii)	correct shape for C (180° phase change) with amplitude < incident correct shape for L (180° phase difference from C) with amplitude < incident amplitude of L equal/slightly less than C (award if first two marks are awarded)	M1 A1 A1																		
																						
	(c)	(i)	<table border="1"><thead><tr><th>refractive index n of lens</th><th>optimum refractive index n_c of the coating</th></tr></thead><tbody><tr><td>1.50</td><td>1.2250</td></tr><tr><td>1.55</td><td>$1.2440 \leq n_c \leq 1.2460$</td></tr><tr><td>1.60</td><td>$1.2640 \leq n_c \leq 1.2650$</td></tr><tr><td>1.65</td><td>$1.2840 \leq n_c \leq 1.2850$</td></tr><tr><td>1.70</td><td>$1.3035 \leq n_c \leq 1.3045$</td></tr></tbody></table> All answers expressed to 4 d.p. (1/2 smallest square)	refractive index n of lens	optimum refractive index n_c of the coating	1.50	1.2250	1.55	$1.2440 \leq n_c \leq 1.2460$	1.60	$1.2640 \leq n_c \leq 1.2650$	1.65	$1.2840 \leq n_c \leq 1.2850$	1.70	$1.3035 \leq n_c \leq 1.3045$	A1						
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		(ii)	$n_c = k\sqrt{n} \Rightarrow k = n_c / \sqrt{n}$ <table border="1"><thead><tr><th>n</th><th>n_c</th><th>k</th></tr></thead><tbody><tr><td>1.50</td><td>1.2250</td><td>1.00021</td></tr><tr><td>1.55</td><td>1.2450</td><td>1.00001</td></tr><tr><td>1.60</td><td>1.2645</td><td>0.99968</td></tr><tr><td>1.65</td><td>1.2845</td><td>0.99998</td></tr><tr><td>1.70</td><td>1.3040</td><td>1.00012</td></tr></tbody></table> use at least three sets of data to verify since k agrees to 3 s.f. (or 4 s.f. depending on data), n_c is proportional to square root of n <i>Only 1 mark awarded if two sets of data used.</i>	n	n_c	k	1.50	1.2250	1.00021	1.55	1.2450	1.00001	1.60	1.2645	0.99968	1.65	1.2845	0.99998	1.70	1.3040	1.00012	M1 A1
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		(iii)	from (ii), $k \sim 1.000$ so, $n_c = \sqrt{1.58} = 1.26(1.257, 1.2570)$	A1																		

	(d)	(i)	visible light comprises of a range of / different wavelengths	M1
			reduce reflection of these wavelengths through <u>different thickness / refractive index</u> of coating (since T and R depends on refractive index)	A1
		(ii)	obtain thickness of one layer by using one wavelength from the of visible light spectrum between 400 nm and 700 nm	M1
			obtain thickness of 7 layers (14 layers if coating both sides considered) with value within the range $0.7 \mu\text{m} \leq \text{thickness} \leq 2.4 \mu\text{m}$	A1
			(minimum = $7 \times \frac{1}{4} \times 400 \times 10^{-9} = 0.7 \mu\text{m}$, maximum = $2 \times 7 \times \frac{1}{4} \times 700 \times 10^{-9} = 2.4 \mu\text{m}$)	
				[Total: 22]